

Crime Exposure & CTA Rail Station Ridership in Chicago

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Introduction

Crime in and around public transit systems is a persistent challenge for urban planners and policy makers. Research has consistently demonstrated that the physical and social environments surrounding transit stations shape both the incidence of crime and riders' perceptions of safety (Loukaitou-Sideris, 1999; Smith & Clarke, 2000). Transit stations can function as both crime attractors i.e. drawing offenders to locations with known opportunities and crime generators where the concentration of large numbers of people at predictable times creates unplanned opportunities for criminal activity (Brantingham & Brantingham, 1993; Smith & Clarke, 2000). The logic is that stations funnel pedestrian traffic through confined spaces at regular intervals, creating a convergence of potential offenders and targets that would not otherwise occur. When station environments also feature poor lighting, limited sightlines, or adjacent land uses such as vacant lots and liquor stores, the conditions for crime are further amplified (Loukaitou-Sideris, 1999). Studies in Los Angeles have shown that crime at bus stops and light-rail stations is highly concentrated spatially and correlates with features of the surrounding built environment such as land use mix, lighting conditions, and proximity to vacant lots (Loukaitou-Sideris, 1999; Loukaitou-Sideris, Liggett, & Iseki, 2002). In Chicago specifically, Block and Block found that street robbery clusters in the environs of rapid transit stations, suggesting that the relationship between transit infrastructure and crime is not merely incidental but spatially structured (2000).

Safety perceptions directly influence transit ridership. When riders feel unsafe at or near stations, they may reduce their use of public transit or avoid certain routes altogether (Hartgen, Ingalls, & Owens, 1993; Loukaitou-Sideris, 2014). A UK Department for Environment, Transport and the Region report estimated that reducing fear of crime on public transit could increase ridership by three percent at peak times and ten percent during off-peak periods (Newton, 2004). In the Chicago context, the CTA rail system, commonly known as the "L", carried approximately 127.5 million rides in 2024, still recovering from pandemic-era declines that brought overall CTA ridership to roughly 68 percent of 2019 levels (CTA, 2024). Recent

high-profile violent incidents on CTA trains and platforms have further amplified public concern about transit safety, suggesting that the relationship between crime and ridership warrants closer spatial investigation.

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Through our analysis, we discovered specific parts of the CTA Rail lines that had lower ridership yet a higher number of crimes committed in 2025 within a 0.5-mile buffer. These insights helped highlight specific stations that are worth further examining to determine the root cause of the low ridership. Our spatial analyses allowed us to conclude that minimum Euclidean distance to a police station did not have a strong correlation to ridership or crime count, discouraging additional police station placement from being a strong solution to any possible relation between crime and low ridership.

Methods

Data definitions

This analysis draws on five datasets, all sourced from the Chicago Data Portal.

- **CTS L station location** (Portal ID: 8pix-ypme). This dataset contains CTA rail stops in Chicago, including station name, MAP_ID (a unique station-level identifier), latitude/longitude coordinates, and line color indicators (Red, Blue, Green, Gray, etc.) Because the dataset lists multiple entries per station, and one per direction or platform, we aggregated records by MAP_ID to produce a single point location per unique station, yielding approximately 145 stations.

- **CTA rail ridership (Annual Totals).** This dataset provides annual ridership figures at the station level, including total rides and average weekday, weekend, and holiday rides. We used the 2025 annual totals for this analysis.
- **Chicago Crime data (2025).** Downloaded via the Chicago data Portal crimes API, this dataset includes over 260,000 reported crime incidents with fields for primary crime type, description, latitude/longitude, date, and arrest status. Each record was geocoded at the block level by the Chicago Police Department.
- **Community area (GeoJSON).** This boundary file contains polygon geometries for Chicago's 77 officially designated community areas, used as a spatial reference layer for mapping
- **Police station locations.** This dataset includes the locations of about 22 Chicago Police Department district stations, with fields for district number, address, and latitude/longitude.

Several steps were taken to clean the dataframes to their end results above. For the CTA rail ridership data, we filtered to obtain only ridership data from 2025 and aggregated the data to reflect yearly totals for each station. We also had to type cast the column attribute recording totals from string to integer. For the crimes dataset, we removed any rows with no location data (longitude and latitude) and geocoded the dataframe. Afterwards, we filtered to only work with crimes of relevant types such as robbery and kidnapping. For the CTA rail station dataframe, we filtered out CTA stations outside of the Chicago community areas. Afterwards, we did a left join with the CTA ridership dataframe, matching for common station ID's. Lastly, for the police station dataframe, entries with no latitude and longitude were filtered out, after which the dataframe was geocoded. Overall, all spatial data were converted to sf objects in R and projected to the Illinois State Plane coordinates reference system, which uses feet as its unit of measurement, enabling distance based buffer calculation.

Spatial analysis

Our analysis proceeded through four primary stages.

- **Buffer zone.** We constructed 0.5-mile (2,640-foot) circular buffer zones around each of the approximately 145 CTA rail stations using the `st_buffer()` function from the `sf` package. The half-mile distance was selected as a reasonable approximation of the

walkable catchment area around a rail station, the zone within which a rider would be exposed to the surrounding environment while traveling to or from the station. A half-mile radius corresponds to roughly a ten-minute walk, which is the standard catchment distance used in transit-oriented development planning and by the Federal Transit Administration for rail stations (Guerra, Cervero, & Tischler, 2012). This threshold captures the environment that a rider would plausibly traverse and perceive as part of their station experience, making it a defensible choice for linking crime exposure to ridership behavior. An alternative approach would be to assign Census blocks to their nearest CTA station, creating Voronoi-style catchment areas that account for the actual service territory of each station rather than imposing a uniform circular buffer. While this method would better reflect spatial variation in station coverage, the buffer approach was selected for its simplicity and interpretability given the project's time constraints.

- **Spatial joining and crime counting.** Using `st_join()`, we performed a point-in-polygon spatial join to identify all crime incidents falling within each station's buffer zone. We then aggregated these results to obtain a crime count per station. To account for the fact that stations with higher ridership tend to be located in denser, higher-activity areas, we also computed a normalized crime rate defined as $(\text{crime count} / \text{total annual ridership}) \times 10,000$. This rate allows for more meaningful comparisons across stations with different levels of activity.
- **Police station proximity analysis.** We examined whether the proximity of a police station to a CTA rail station was associated with differences in nearby crime levels. First, we applied a binary classification: stations with at least one police station within their 0.5-mile buffer were classified as "police station nearby," and the remainder as "no police station nearby." We then mapped crime exposure levels separately for each group. As a complementary approach, we calculated the Euclidean distance from each CTA station to its nearest police station using `st_nearest_feature()` and `st_distance()` from the `sf` package, and compared these distances against crime rate patterns.
- **Non spatial exploratory.** We produced scatter plots of station-level ridership against crime count within the 0.5-mile buffer and fitted a simple ordinary least squares (OLS) regression to assess the linear relationship. Summary statistics including R-squared,

slope, and p-value were computed to characterize the strength and direction of the association.

All analyses were conducted in R (version 4.x) using RStudio. Key packages included *sf* for spatial data handling, buffer creation, and spatial joins; *tmap* for cartographic visualization and choropleth mapping; and *dplyr* for data wrangling and aggregation. Used packages were cited in the references below.

Designing process

We developed our cartographic products iteratively, beginning with simple point maps of CTA stations and police stations overlaid on community area boundaries to build spatial intuition. Buffer maps were rendered using graduated color symbology with five categories (Very Low, Low, Moderate, High, Very High) determined by quantile classification to explore general distribution. All maps used a sequential color scheme except for three. Those three utilized a viridis or red-blue color palette to remain accessible for color blindness. Crime count and crime rate maps were presented side by side to highlight the distinction between absolute and normalized measures of crime exposure. The police station analysis maps used the same color scheme for consistency. To maximize readability and contextual intuition, maps included scale bars, legends, and descriptive titles. The scatter plot was annotated with regression statistics (R^2 , slope, p-value) directly on the figure for immediate interpretability.

Results

Crime Buffer Analysis and Ridership

We plotted CTA Rail ridership totals by station as seen in Figure 1. The results were that the highest totals in ridership came from stations located around Downtown Chicago. Rail lines extended outwards, with stations gradually decreasing in ridership totals the further away from the city center. Ridership remained relatively higher for the stations part of the Blue line from O'Hare, and relatively lower for the Blue, Pink, and Green lines extending horizontally from Western Chicago. South of Downtown Chicago, the Red and Green lines can also be seen having lower levels of ridership.

Figure 2 shows the results of counting the number of relevant crimes committed in a 0.5-mile buffer around each CTA Rail station. The highest totals were in the buffers around CTA

Rail stations in the city center. The Blue line featuring O'Hare and the Orange line consistently had low counts of crime, while the opposite was the case for the Red, Green, Pink, and horizontal Blue lines. Figure 3 shows our attempt to normalize the data, showing the 'crime rate,' calculated by dividing the crime count in the 0.5-mile buffer by the ridership total for the given station. The visual results remain similar between Figure 2 and Figure 3.

There are two patterns that emerge from comparing Figure 1 to Figure 2. We did not compare Figure 3, as ridership count was directly worked into 'crime rate.' The first pattern was that in the city center, high counts of crime and high ridership coincide. The second pattern was that excluding the city center, higher levels of crime did seem to be related to lower ridership totals. For example, the stations of the Red and Green line in Southern Chicago with the lowest totals of ridership were also the stations that had the highest counts of crime for those lines. Similarly, lines moving horizontal from Western Chicago also had lower ridership totals. The opposite also seemed to be true—the Blue line from O'Hare and the Orange line were two examples of relatively high ridership and low crime.

Figure 4 shows the non-spatial correlation between crime count and ridership (in millions) through a scatter plot. There was a positive correlation with a slope of 274.904, accompanied by an extremely small p-value of $2.78e^{-12}$ with a moderately-low R^2 -coefficient of 0.329.

Police Stations

Viewing the results from our police station analysis, we found a few interesting conclusions. Looking at the map of crime exposure for stations with a police station nearby (Figure 5), and comparing it to the map of stations without a police station nearby (Figure 6), we see that there is not that much of a difference in the distribution of crime exposure between the 2 maps. There seems to be a similar number of stations in each crime level category for both of the maps. In fact, stations without a nearby police station seem to have more very low-crime stations and fewer very high-crime stations. This is likely due to the confounding factor that the city chooses the locations of its police stations based on crime rates, meaning low-crime areas probably won't have a high density of police stations. We see similar results when viewing CTA stations by minimum distance to a police station (Figure 7), as many of the highest distances to police stations among CTA rail stations also have very low crime rates. A scatterplot comparing

crime rate to minimum distance from any police station (Figure 8) also shows this correlation. Figure 8 shows a slight negative correlation between the two variables, with a slope of -4.381, p-value of 0.202, and a R^2 -coefficient of 0.013.

Discussion

Summary of Findings

Our analysis reveals a spatial pattern that both confirms and complicates the initial hypothesis. Crime is not uniformly distributed across CTA rail stations; rather, it concentrates heavily in the downtown Loop area and along certain segments of the Red, Green, and Blue lines. Stations on the South and West sides also show elevated crime rates when ridership is taken into account through the normalized crime rate measure. This distinction between absolute crime counts and crime rates proved analytically important: downtown stations exhibit the highest raw crime counts, but because they also have the highest ridership, their crime rates are not necessarily the most extreme. Conversely, lower-ridership stations on the South Side show disproportionately high crime rates, suggesting that riders at those stations face greater per-capita crime exposure.

The central finding from the regression analysis is that crime count and ridership are positively correlated. This is the opposite direction from our initial hypothesis, which predicted that higher crime would be associated with lower ridership. However, this positive relationship is most parsimoniously explained by a common underlying factor: population density and urban activity levels. Stations located in dense, high-traffic areas naturally attract both more riders and more crime, not because crime increases ridership but because both are driven by the volume of people present. This confounding effect was anticipated and is consistent with findings from Loukaitou-Sideris et al. (2002), who noted that property crimes are more prevalent in neighborhoods with higher population density and that violent crimes tend to occur at busier stations.

The police station proximity analysis yielded a null result: there is no discernible difference in crime exposure between CTA stations with a nearby police station and those without one. The minimum-distance analysis further confirmed this, stations closer to police stations do not systematically have lower crime. This likely reflects the fact that police stations

are sited in response to existing crime patterns rather than exerting a strong deterrent effect at the spatial scale of our analysis.

Limitations

Several limitations should be noted. First, our analysis does not control for population density, employment centers, income levels, or other socioeconomic variables that are likely to confound the relationship between crime and ridership. Without these controls, we cannot disentangle the degree to which crime independently affects ridership from the degree to which both variables simply reflect urban density. Second, there is a temporal mismatch in our data: ridership is aggregated annually while crime data captures individual incidents throughout the year. Third, our buffer zones impose a uniform circular catchment around each station, which doesn't account for differences in street network connectivity, pedestrian access, or physical barriers that shape actual walking routes to stations. Fourth, we treat all crime types equally in our counts but not all crime types are equally relevant to the question of whether crime deters ridership. Person to person crimes like assault, robbery, and battery are likely to have a stronger effect on safety perceptions than property crimes like motor vehicle theft or criminal damage, which may not directly affect a rider's sense of personal safety at a station. Finally, this analysis is exploratory and correlational, meaning we make no causal claims about whether crime deters ridership or whether other mechanisms are at work.

Future Work

There are a few things we may want to consider in the future to get more conclusive results. One would be to consider the population in our buffer analysis. Population density likely has an impact on both crime and CTA ridership, and using population along with crime in our buffer analysis could help us understand more about how influential crime is on ridership. Additionally, we could consider using a different analysis method, such as assigning census blocks to the nearest CTA station, instead of buffers, and comparing the results. We may also want to filter out non-person-to-person crimes in the future, as they likely wouldn't affect ridership.

Appendix

Responsibilities:

- Andy Pincus: Results (Police Stations), Discussion (Future Work)
- Blair Peng: Introduction, Methods, Discussion (Summary of Findings and Limitations)
- Julien Lee: Introduction, Methods, Results (Crime Buffer Analysis and Ridership)

CTA Rail Station Ridership (2025)

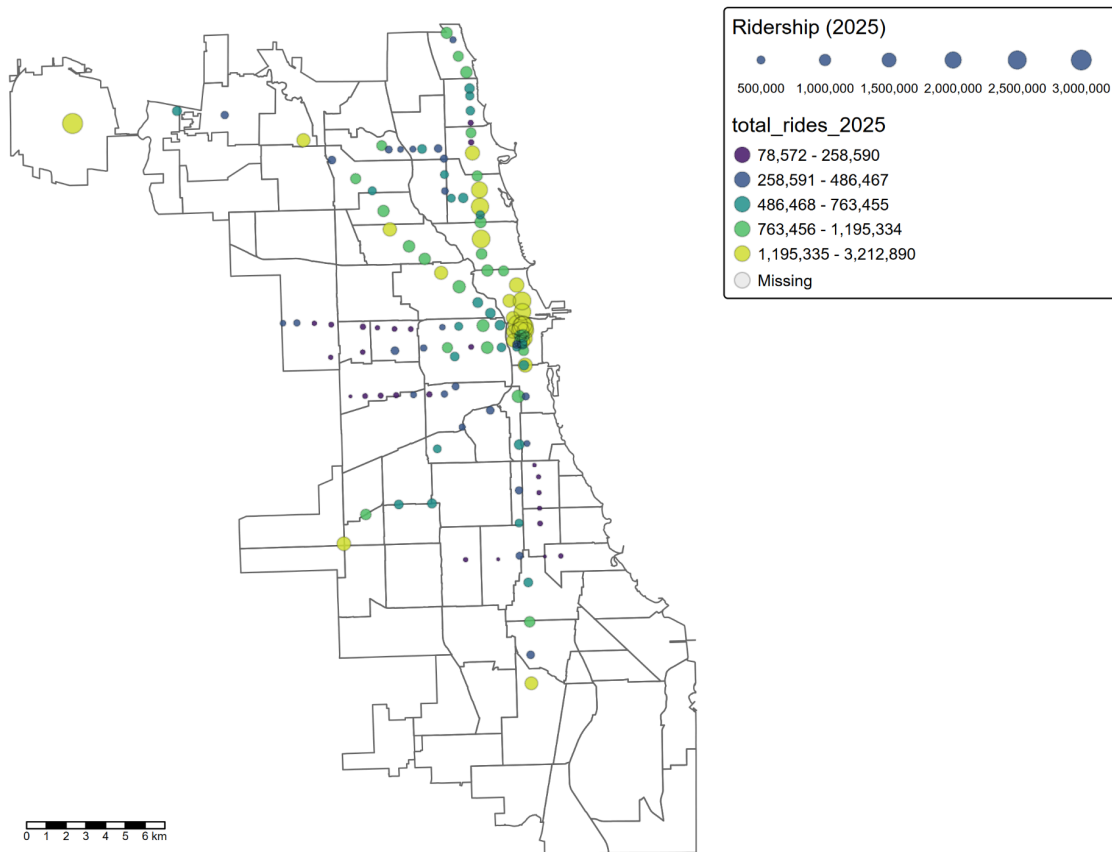


Figure 1. CTA Rail Station Ridership (2025). Graduated point map showing total annual rides by station, with point size and color scaled by ridership volume. Stations are overlaid on Chicago community area boundaries.

CTA Rail Stations Crime Count: 0.5-mile Buffer

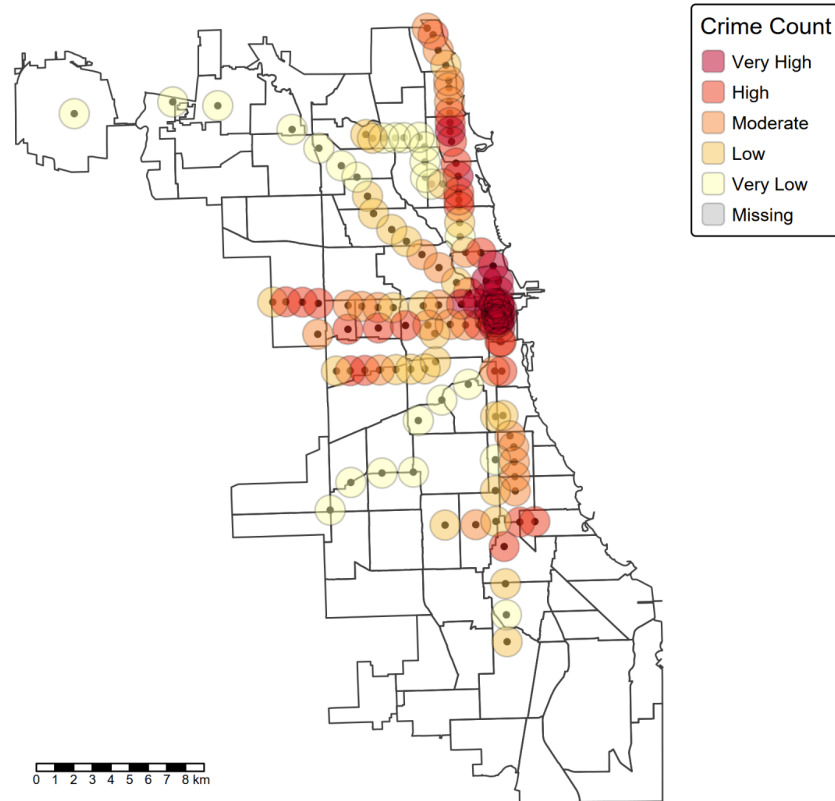


Figure 2. CTA Rail Stations Crime Count: 0.5-mile Buffer. Graduated color map showing the total number of reported crime incidents (2025) within a half-mile buffer of each CTA rail station, classified into five categories using natural breaks.

CTA Rail Stations Crime Rate: 0.5-mile Buffer

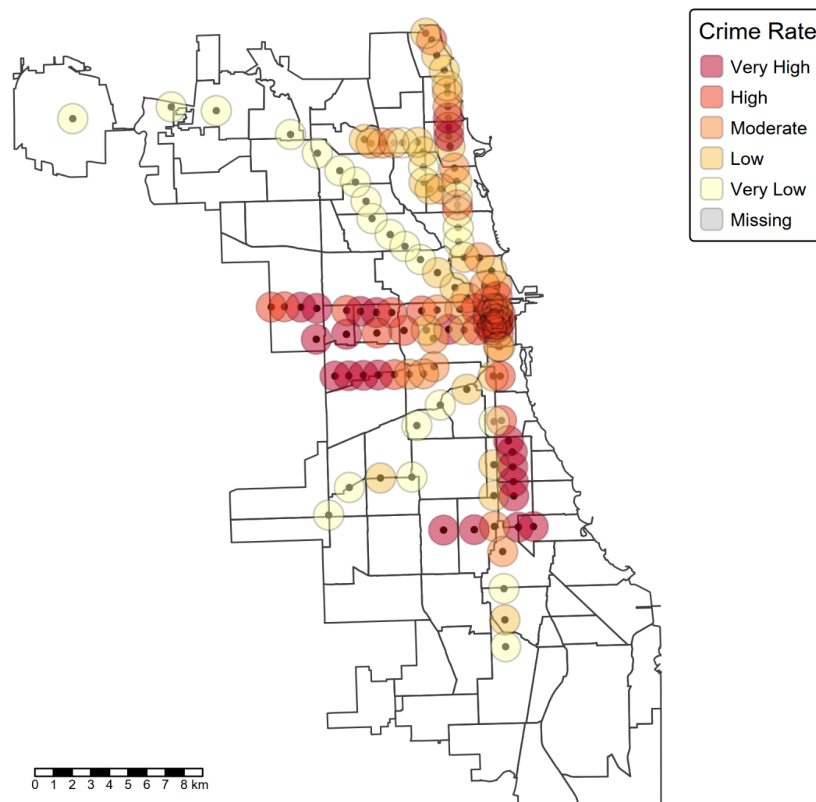


Figure 3. CTATail Stations Crime Rate: 0.5-mile Buffer, Normalized crime rate per station, calculated as $(\text{crime count}/\text{total annual ridership}) \times 10,000$, within a half-mile buffer. This measure adjusts for differences in station activity levels.

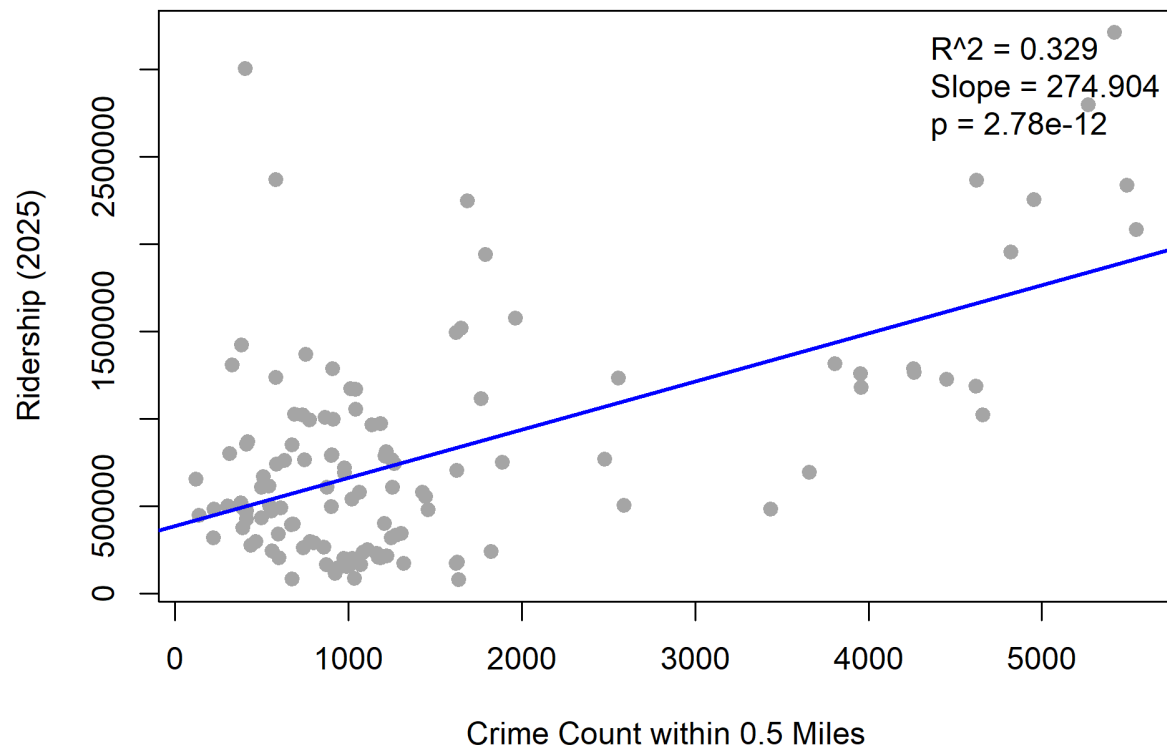
CTA 2025 Ridership vs Crime Count

Figure 4. STA 2025 Ridership vs. Crime Count (0.5-mile Buffer). Scatter plot of station-level annual ridership against crime count within half a mile, with OLS regression line ($R^2 = 0.329$, slope = 274.904, $p=2.78e-12$).

CTA Rail Stations Crime Exposure: 0.5-mile Buffer with a Police Station Nearby

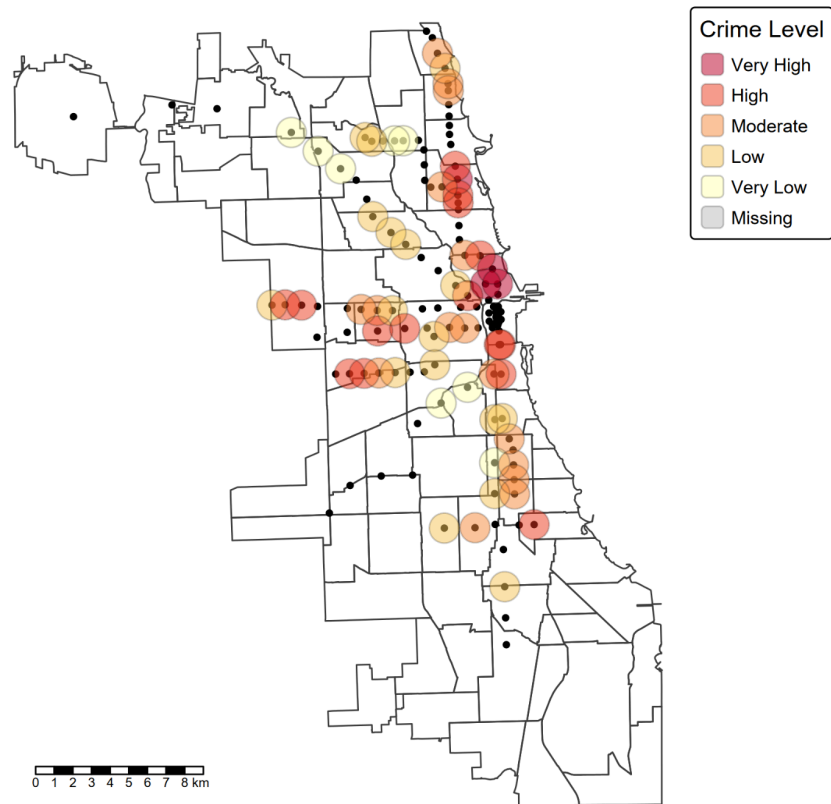


Figure 5. CTA Rail Stations Crime Exposure: 0.5-mile Buffer with a Police Station Nearby. Subset of stations that have at least one Chicago Police Department district station within their half-mile buffer, colored by crime level. Black dots indicate stations without buffer crime data (missing ridership).

CTA Rail Stations Crime Exposure: 0.5-mile Buffer without a Police Station Nearby

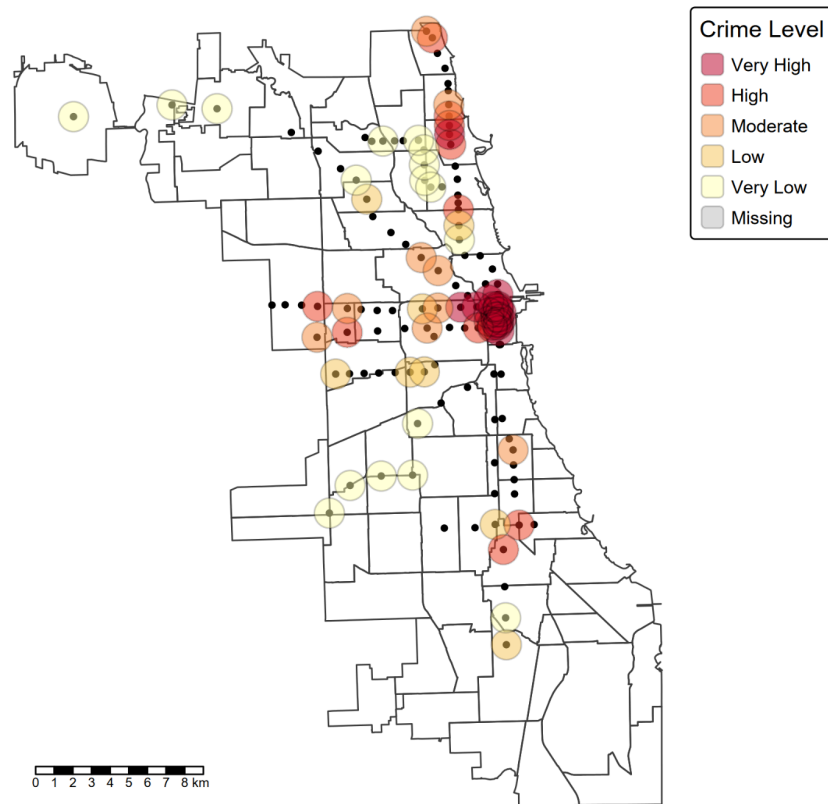


Figure 6. CTA Rail Stations Crime Exposure: 0.5-mile Buffer without a Police Station Nearby. Subset of stations that do not have a Chicago Police Department district station within their half-mile buffer, colored by crime level. Compared with Figure 5, the distribution of crime levels is similar, suggesting police station proximity has limited association with nearby crime.

CTA Stations: Distance to Nearest Police Station

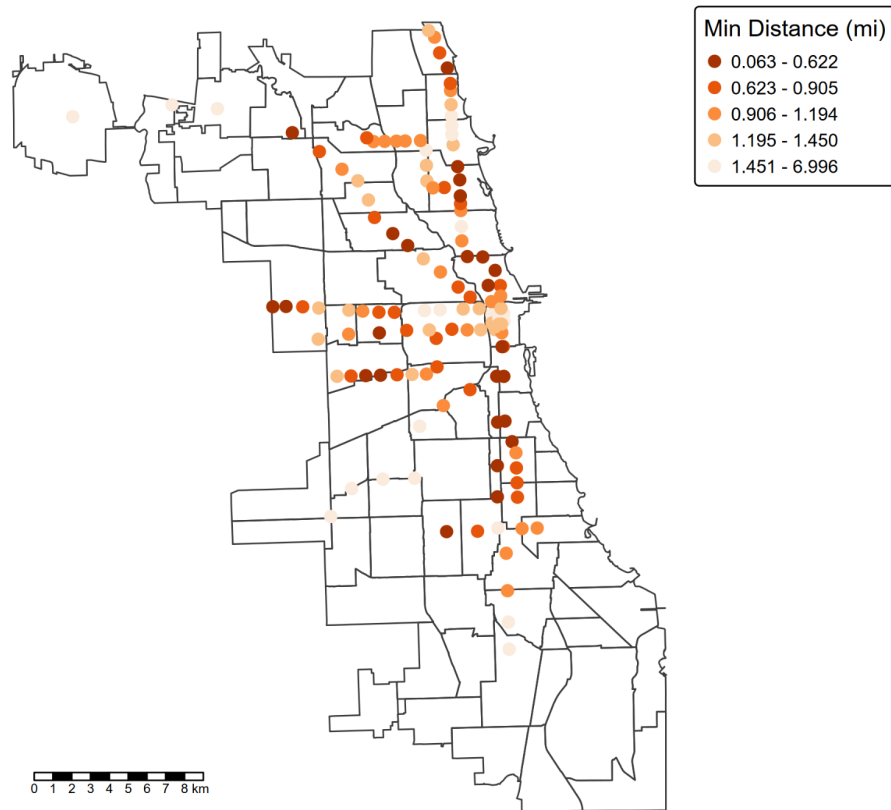


Figure 7. CTA Stations: Distance to Nearest Police Station. Euclidean distance (in miles) from each CTA rail station to its nearest Chicago Police Department district station, classified into five categories. Darker points indicate closer proximity. Stations in the city center tend to be closer to police stations, while outlying stations on the South and far Northwest sides are farthest.

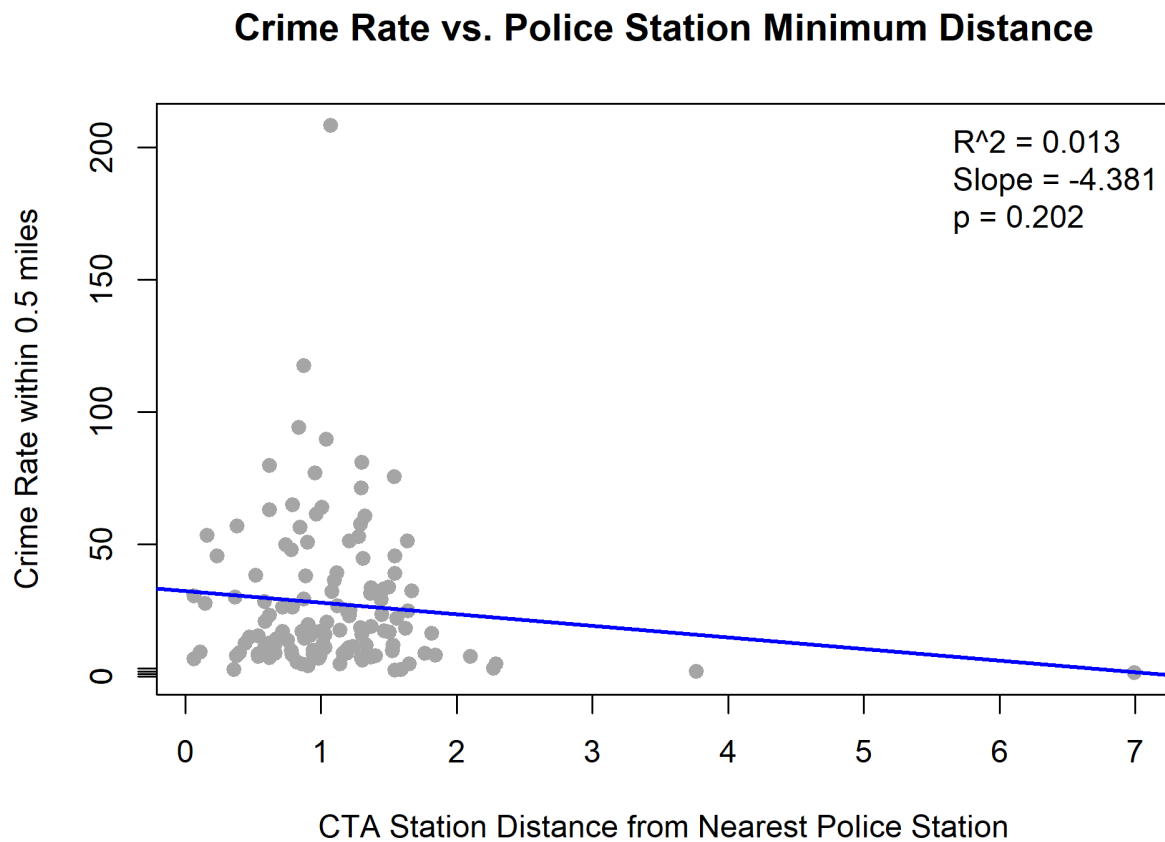


Figure 8. Crime Rate vs. Police Station Minimum Distance. Scatter point of normalized crime rate (per 10,000 riders) against distance to nearest police station (miles), with OLS regression line ($R^2 = 0.013$, slope = -4.381 , $p = 0.202$). The near zero R^2 and non-significant p -value indicate no meaningful linear relationship between police station proximity and crime rate.

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